

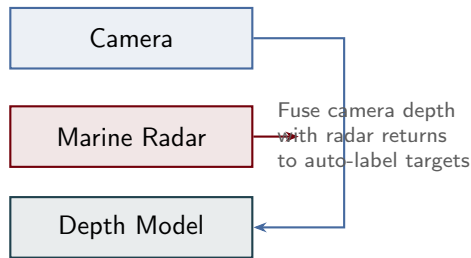
Why Depth Matters

Knowing 3D from 2D enables:

- ▶ **Autonomous driving** — obstacle distance, lane geometry
- ▶ **Robotics** — grasping, navigation, collision avoidance
- ▶ **AR / VR** — object occlusion, scene placement
- ▶ **3D reconstruction** — single-image scene modeling

All require *metric* depth — real-world distances in meters, not just relative ordering.

My use case: marine radar labeling



Need metric depth at 50–500 m over open water.

Problem: Most depth models fail at long range and on textureless surfaces like water.